

SMART: A Selective Marker-guided Adaptive Recursive Transformer with Learned Gene-Graph Routing for Single-Cell and Multi-Omics Classification

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Abstract

Transformer foundation models for single-cell transcriptomics treat every one of the thousands of measured genes as an equally important token and stack many independent layers, which makes them parameter-heavy and leaves the biology of *which* genes deserve computation entirely to be learned from data. We argue that for gene expression, parameter efficiency should be an *architectural* property, and that biology should enter the *routing decision* rather than only a post-hoc interpretation step. We present **SMART** (Selective Marker-guided Adaptive Recursive Transformer), which (i) learns end-to-end which genes are *markers* worth dedicated computation through a cross-attention *marker router* whose learnable queries attend over all genes; (ii) represents each cell by only its $M \ll N$ markers, cutting self-attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$; and (iii) processes the marker tokens with a *single* transformer block applied recursively, where a Mixture-of-Recursions router grants each gene its own adaptive recursion depth, so depth becomes an intrinsic compute-allocation importance score. The same interface extends beyond expression: for bulk multi-omics the tokens are fixed *Reactome pathway* tokens that pool a pathway’s mutation, copy-number and expression channels. On top of the tokens we study a **biology-informed router** that folds a gene-gene network-centrality prior into the depth decision, so hub genes/pathways recurse deeper, and we contrast four ways of forming that graph under otherwise identical training: a *fixed* co-expression / Reactome centrality, a degree-matched *random-graph* control, a *learned* routing graph trained end-to-end from random init, and that learned graph *warm-started* from the biological graph. Our controlled study spans 11 datasets (8 genomap single-cell suites and 3 Reactome/P-NET multi-omics cohorts) and yields one positive headline alongside honest negatives. **Positive:** the *learned* routing graph raises single-cell macro-F1 to 70.8% (+11.7 points over a no-prior router and well above a degree-matched random graph, 57.5%), the decisive routing result; warm-starting it from biology neither helps nor hurts materially, so the model recovers the useful structure from data on its own. **Negatives:** a *fixed* hand-built biology prior does *not* help, it sits at or below its random-graph control (42.4% vs. 57.5% on single cell) and col-

lapses on several suites, and no routing variant improves calibration (NLL/ECE) – though a standard post-hoc temperature scaling sharply cuts the log-likelihood error of every model at *no* accuracy cost. Alongside these, two efficiency properties hold: *parameter reduction* (one shared block uses $1/K$ of the parameters of K independent layers, a $4\times$ reduction at $K=4$, at comparable accuracy) and *token reduction* (a few dozen to a few hundred interpretable tokens recover most full-gene accuracy), and adaptive depth buys a $\sim 38\%$ *compute* saving at matched accuracy. We report the controlled comparison exactly as the runs deliver it. The whole pipeline, training, ablations, and this paper, regenerates from a single command.

1 Introduction

Single-cell RNA sequencing now profiles the expression of thousands of genes across millions of cells, and a wave of transformer *foundation models*, scGPT (Cui et al. 2024), Geneformer (Theodoris et al. 2023), scBERT (Yang et al. 2022) and scFoundation (Hao et al. 2024), has adapted the architecture of (Vaswani et al. 2017) to this modality. These models are powerful but inherit two costly habits from language transformers. First, they treat *every* gene as an equally important token, so a housekeeping gene and a lineage-defining marker get identical computational budgets. Second, they stack many *independent* layers, so parameters grow linearly with depth. The result is models with tens to hundreds of millions of parameters (Hao et al. 2024) whose self-attention scales quadratically in the number of genes, and whose efficiency is usually recovered only afterward through pruning or distillation.

We take a different stance: *for gene expression, the data and the known biology together tell us where to spend computation*. Decades of single-cell biology rest on two facts. A small set of *marker genes* is sufficient to discriminate cell types (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023); and gene co-expression and regulatory networks are approximately scale-free, so a few high-degree *hub* genes exert outsized influence. If a model could decide, during training, which genes are markers, and could be told, before training, which genes are network hubs, it could grant those genes dedicated capacity and let everything else share parameters. This makes

parameter efficiency an *architectural* property and lets biology shape the *routing decision* rather than merely validate it afterward.

We realise this in **SMART**, a recursive marker-guided transformer with three coupled components and one biological prior. (1) A cross-attention *marker router*: M learnable queries attend over all N genes with a temperature-annealed softmax (Set-Transformer / Perceiver-style (Jang, Gu, and Poole 2017; Baln, Abid, and Zou 2019)), so the model learns *which* genes are markers end-to-end while gradients reach every gene. (2) *Marker-driven compression*: each cell is represented by only its $M \ll N$ markers, cutting attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$. (3) A *recursive shared block*: one transformer block applied K times in the spirit of Universal Transformers (Dehghani et al. 2019), ALBERT (Lan et al. 2020) and Mixture-of-Recursions (Bae et al. 2025), with an expert-choice depth router that gives each gene an adaptive recursion depth. On top of this, a *biology-informed router* folds a gene-gene interaction graph into that depth decision. Our central finding concerns *how* that graph should be formed: a *fixed* hand-built graph (genomap co-expression (Islam and Xing 2023) or Reactome centrality) does not beat a degree-matched random-graph control, whereas a *learned*, data-driven graph trained end-to-end does – so biology helps routing only when the graph is learned rather than imposed a priori.

We make the following contributions:

- We propose a **biology-informed recursion router** that folds a gene-gene graph into the depth-routing decision, so biology shapes *where compute goes* rather than only how results are interpreted. The graph can be a fixed network-centrality prior (an annealed additive bias) or, in our best variant, a low-rank graph *learned end-to-end*; we give the mechanism a full mathematical and empirical-Bayes grounding.
- We evaluate the router with a controlled *none / random-graph / fixed-biology / learned* ablation and find the **learned** routing graph is the decisive positive: it lifts single-cell macro-F1 to 70.8% (+11.7 points over a no-prior router), whereas a fixed hand-built biology prior does not separate from a degree-matched random-graph control, isolating what actually helps from generic bias.
- We propose SMART, a transformer for genomic classification in which token selection, token compression, and parameter-shared recursion are co-designed and trained end-to-end, with each token’s recursion depth serving as an intrinsic compute-allocation importance score. The token interface is interpretable and modality-general: *learned marker genes* for single-cell expression, and fixed *Reactome pathway tokens* (pooling each pathway’s mutation, copy-number and expression channels) for bulk multi-omics.
- We run a controlled study of the full recursive-transformer design space – token count, marker vs pathway tokens, weight sharing, expert- vs token-choice routing, and the routing prior – summarised in *five* result tables. Two are deliberate negatives we surface rather than bury: a fixed hand-built biology prior *neither beats a vanilla transformer on accuracy* (Table 1) *nor improves its calibration* (NLL/ECE,

Table 5). The others are the wins: **learned routing** (Table 1), **parameter reduction** (one shared block uses $1/K$ of the parameters of K independent layers, a $4\times$ reduction at $K=4$; Table 3), **token reduction** (a few dozen to a few hundred tokens recover most full-gene accuracy; Table 4), and an adaptive-depth **compute** saving ($\sim 38\%$ fewer recursion FLOPs at matched accuracy; Table 2).

- We evaluate on all 11 datasets – 8 genomap single-cell suites (Baron, Lung, Muraro, Oesophagus, Segerstolpe, Spleen, T-cell, Xin) and 3 Reactome/P-NET multi-omics cohorts (prostate, BLCA, STAD), with no TCGA bulk data – showing the same design decisions transfer across modalities; all ablations report multi-seed mean $\pm$ std.
- We release a fully reproducible pipeline in which a single command runs all experiments and regenerates this paper, numbers, tables and figures included.

2 Related Work

Transformer foundation models for single-cell omics. Geneformer (Theodoris et al. 2023) and scBERT (Yang et al. 2022) adapt masked-language-model pretraining to single-cell transcriptomes; scGPT (Cui et al. 2024) scales generative pretraining to 33M cells; scFoundation (Hao et al. 2024) trains a 100M-parameter model over $\sim 20,000$ genes; CellPLM (Wen et al. 2024) and Cell2Sentence (Levine et al. 2024) push cell- and language-level pretraining further. Earlier deep generative approaches such as scVI (Lopez et al. 2018) established probabilistic latent representations. genomap (Islam and Xing 2023) instead reshapes the gene vector into an image via an optimal-transport layout built from a gene-gene *interaction* matrix, and uses a small CNN (genoNet) for cell recognition; we reuse precisely that interaction identification, but as a routing prior rather than an image layout. All of these treat genes uniformly or select them in a separate stage; none make marker-driven sparsity an architectural prior or let a gene-interaction graph shape adaptive computation.

Parameter-efficient and recursive transformers. Tying weights across depth was shown to retain representational power by Universal Transformers (Dehghani et al. 2019) and to shrink models in ALBERT (Lan et al. 2020). Mixture-of-Recursions (Bae et al. 2025) unifies weight sharing with token-level adaptive depth, and Mixture-of-Depths (Raposo et al. 2024) routes tokens through variable numbers of layers, both echoing sparsely-gated mixtures of experts (Shazeer et al. 2017) and adaptive computation time (Graves 2016). Efficient-attention methods, Linformer (Wang et al. 2020), Performer (Choromanski et al. 2021) and Nystromformer (Xiong et al. 2021), reduce the quadratic cost generically. Recursive weight sharing has also been pushed in vision and restoration transformers: the Sliced Recursive Transformer (Shen, Liu, and Xing 2022), RISTRA (Zhou et al. 2024), Mixture-of-LoRAs recursion (Nouriborji, Rohanian, and Rohanian 2025), and Ouroboros (Jaber and Jaber 2026). We borrow the weight-sharing mechanism but make the token set biologically structured and the routing biologically primed,

so our $\mathcal{O}(M^2)$ saving and our prior are complementary to these methods.

Markers, networks, and biological priors. Marker-based annotation tools such as scType (Ianevski, Giri, and Aittokallio 2022), Cell-ID (Cortal et al. 2021) and SingleR (Aran et al. 2019), and curated databases including PanglaoDB (Franzén, Gan, and Björkegren 2019) and CellMarker 2.0 (Hu et al. 2023), encode the principle that few genes carry most discriminative signal. Pathway-informed models such as the graph transformer PATH (Howlader, Islam, and Le 2026) build structure from Reactome (Gillespie et al. 2022). Most prior work uses such resources to *validate* learned features; we instead move a label-free network prior *into* the routing decision. Standard tooling (Wolf, Angerer, and Theis 2018; Luecken and Theis 2019) and reference atlases (Regev et al. 2017; The Tabula Sapiens Consortium 2022) provide the broader context.

3 Method

Token interface (markers and pathways). SMART turns the input into $M \ll N$ interpretable tokens before any quadratic attention. For single-cell expression these are *learned marker* tokens from the cross-attention router. For bulk multi-omics they are fixed *Reactome pathway* tokens: a sparse gene→pathway membership pools each pathway’s per-gene channels (mean pooling for dense assays such as copy-number and expression; burden/sum pooling for sparse binary mutation), so every token is a named pathway. Both feed the same recursive stack, and on top of token selection we study the full design space – token count, weight-sharing scheme (Cycle / Sequence / Middle-Cycle / Middle-Sequence, from one shared block to K independent ones), expert- vs token-choice depth routing, reuse of the first-step attention key/value across recursions, and warm-starting the shared block from a fixed-depth model.

Overview

Let $x \in \mathbb{R}^N$ be the expression vector of a cell over N genes. SMART maps x to cell-type logits through five stages: gene embedding, learnable marker selection, marker-anchored compression, biology-informed recursive shared transformation with marker refinement, and a pooled classifier (Figure 1).

Gene Embedding

Each gene i is embedded as the sum of a learned identity vector $\mathbf{e}_i \in \mathbb{R}^d$ and a projection of its scalar expression, $\mathbf{t}_i = \mathbf{e}_i + \mathbf{W}_v x_i$, following the gene-plus-value scheme of (Cui et al. 2024; Theodoris et al. 2023). This is linear in N and runs over all genes before any compression.

Cross-Attention Marker Router

We identify markers with a cross-attention *router*. We maintain M learnable marker queries $\mathbf{q}_m \in \mathbb{R}^d$; each attends over the genes through a shared key projection $\mathbf{k}_i = \mathbf{W}_k \mathbf{e}_i$, giving selection weights $\mathbf{w}_m = \text{softmax}(\mathbf{q}_m \mathbf{K}^\top / (\tau \sqrt{d}))$ over *all* N genes, with temperature τ annealed from soft

to peaked. Because the softmax spans all genes, gradients reach every gene, so a query can migrate to an informative gene it did not initially favour, the property hard top- k routing lacks. Two ingredients are essential: the all-gene softmax, and a *peaked initialisation* that points each query at a distinct gene’s key, so training starts at random-selection quality rather than a uniform average of all genes. At inference each query collapses to its arg-max gene $g_m = \arg \max_i \mathbf{q}_m^\top \mathbf{k}_i$, giving discrete, interpretable markers and $\mathcal{O}(M^2 d)$ attention. As alternatives we consider the Concrete selector (Baln, Abid, and Zou 2019; Jang, Gu, and Poole 2017) and fixed variance- and random-selected panels.

Marker Tokens

Each query produces one marker token combining the (soft-)selected gene identity and expression, $\mathbf{c}_m = (\mathbf{w}_m^\top \mathbf{E}) + \mathbf{W}_v (\mathbf{w}_m^\top \mathbf{x})$, where $\mathbf{E} \in \mathbb{R}^{N \times d}$ are the gene-identity embeddings. The two contributions are placed on a common scale by the pre-norm LayerNorm at the entry of the shared block (Sec. 3).

Recursive Shared Transformer with Marker Refinement

Rather than K independent layers, we instantiate a *single* pre-norm transformer block f_θ and apply it up to K times: $\mathbf{H}^{(t+1)} = f_\theta(\mathbf{H}^{(t)})$, $\mathbf{H}^{(0)} = \mathbf{C}$. This ties all depth-wise parameters (Dehghani et al. 2019; Lan et al. 2020; Bae et al. 2025), so the stack’s parameter count is independent of K . After each pass we recompute a per-token gate from the *updated* embeddings, $g_m^{(t)} = \sigma(\text{MLP}_r(\mathbf{H}_m^{(t)}))$, and apply it before the next pass (*recursive marker refinement*).

Mixture-of-Recursions over genes. Spending the full depth K on every marker is wasteful: most genes are settled after one pass, while a few lineage drivers reward deeper iteration. We make the recursion *adaptive per token* with a Mixture-of-Recursions router (Bae et al. 2025) (Figure 2). Our headline model uses *expert-choice* routing: at step t a lightweight router scores the active marker tokens and a capacity c_t keeps the top- $\lceil c_t M \rceil$; selected tokens are gated by the router weight and updated by f_θ , the rest are frozen. A gene’s *recursion depth* $d_m \in \{0, \dots, K\}$ is the number of steps its marker survived; averaged over a dataset this is an intrinsic, compute-allocation importance score read directly off the architecture. As an ablation we implement *token-choice* routing (Bae et al. 2025) with a Switch-style load-balancing loss (Shazeer et al. 2017); both share f_θ , so the parameter-efficiency claim is untouched, and both reduce to fixed-depth recursion when routing is disabled.

Biology-Informed Routing

This is the core of our method. So far the router decides depth from data alone, and biology enters only afterward, when we cross-check deeply-routed genes against known markers. We instead move a biological prior *into* the routing decision. For

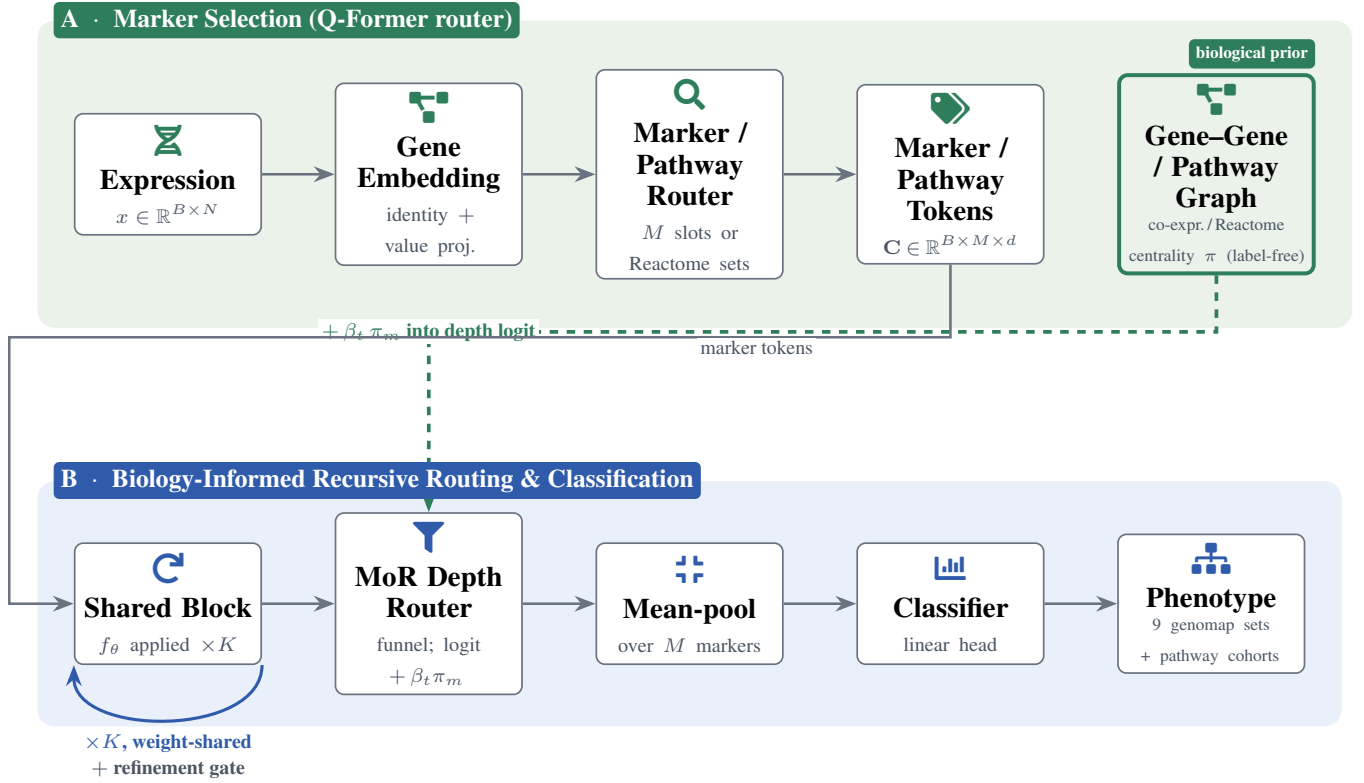


Figure 1: **System overview.** **Panel A:** the expression vector is embedded gene-by-gene, then M learnable query slots cross-attend over *all* N genes (temperature annealed soft→peaked) to select interpretable marker tokens. **Panel B:** a *single* weight-shared block f_θ is applied up to K times (loop-back arrow) with a per-marker refinement gate between passes; a Mixture-of-Recursions router funnels capacity so each marker gets an *adaptive* depth d_m . **Biology-informed routing (our centerpiece):** a genomap gene-gene co-expression graph supplies a label-free network-centrality prior π that is added (annealed by β_t) to the depth-router logit (dashed arrow), so co-expression hub genes get a head start in the funnel without any label leakage. Tokens are mean-pooled and classified; the *same* pipeline serves both single-cell datasets.

marker token m at step t , the expert-choice logit becomes

$$\tilde{r}_m^{(t)} = \underbrace{\frac{1}{\tau} \mathbf{w}_r^\top \mathbf{H}_m^{(t)}}_{\text{data-driven (learned)}} + \underbrace{\beta_t \pi_m}_{\text{biological prior}}, \quad (1)$$

and the keep/drop top- $\lceil c_t M \rceil$ and the gate $g_m^{(t)} = \sigma(\tilde{r}_m^{(t)})$ run exactly as before, so the recursion-depth definition is unchanged but now reflects prior and data together. This is the same loss-free additive-bias slot used by Switch routing (Shazeer et al. 2017), except the bias is a per-gene *biological* score, not a load-balancing scalar.

The prior π_m from gene-gene interactions. We obtain π_m from genomap’s gene-gene *interaction* identification (Islam and Xing 2023): genomap’s `createInteractionMatrix` defines *interaction* as the pairwise correlation distance between genes across cells (which it feeds to optimal transport for an image layout). We call that same function on the training split and reuse only its interaction matrix, taking the co-expression affinity $\mathbf{W}_{ij} = |\text{corr}(g_i, g_j)| = |1 - d_{ij}|$, sparsifying it to each gene’s k nearest neighbours, symmetrising, and reading off

the network centrality

$$\pi = \text{zscore}(\text{eigvec-centrality}(\mathbf{W})), \quad (2)$$

so co-expression *hub* genes, whose perturbation propagates widely, receive a larger prior and are nudged to recurse deeper. Crucially $\mathbf{W}$ is built from *expression alone*, with *no labels*, so the prior injects biological network structure without leaking the cell-type labels; this is what keeps any gene-discovery claim honest.

Annealing β_t . A fixed strong prior would behave like hard routing, pinning compute to known hubs and unable to discover new genes. We therefore warm-start: $\beta_t = \beta_0(1 - \text{progress})$ decays to 0 over training, so the prior dominates early, when hidden states are still random, and the data-driven term takes over late. This is empirical-Bayes shrinkage: a strong prior when evidence is weak, fading as evidence accumulates. Because π_m is a constant additive bias, $\tilde{r}_m^{(t)}$ stays smooth in $\mathbf{w}_r$ and the gate carries gradient, so trainability and the parameter-efficiency claim are unchanged. We validate the component against a degree-matched *random*-graph control: only if the real co-expression graph beats

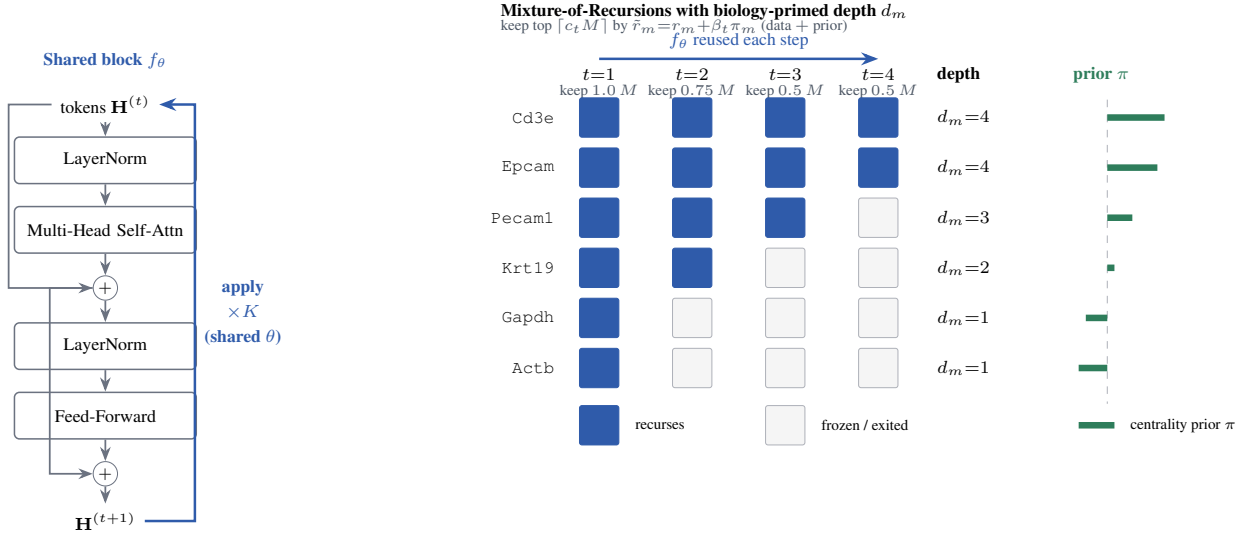


Figure 2: **Biology-informed Mixture-of-Recursions.** *Left:* one weight-shared pre-norm block f_θ (the model’s only transformer parameters) is re-applied up to $K=4$ times, so depth costs no extra parameters. *Right:* an expert-choice router keeps a shrinking top fraction of markers per step (capacity funnel $1, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}$); a marker not kept is frozen, so its *recursion depth* d_m is the deepest step it survived. The keep decision adds a label-free genomap co-expression-centrality prior $\beta_t \pi_m$ to each logit (Eq. 1); the **green bars** show that prior π (longer = more central a co-expression hub), so lineage and hub genes (Cd3e, Epcam) are primed to recur deepest while settled house-keeping genes (Gapdh, Actb) exit at $d_m=1$. The bar heights are illustrative of the centrality ordering, not fitted values.

the random one is it the biology, not mere smoothing, that helps (Sec. 4).

The learned graph (our best variant). A fixed graph is frozen and label-free; it cannot adapt to the task. We therefore also let the model *learn* its own gene-gene graph. We attach a low-rank gene embedding $\mathbf{E} \in \mathbb{R}^{N \times r}$ ($r=16$) whose row-cosine similarity is the affinity, $\mathbf{A} = \tilde{\mathbf{E}}\tilde{\mathbf{E}}^\top$ with $\tilde{\mathbf{E}} = \text{normalize}(\mathbf{E})$, and smooth the input along it before marker selection, $\mathbf{x} \leftarrow (1 - \lambda)\mathbf{x} + \lambda(\mathbf{x}\tilde{\mathbf{E}})\tilde{\mathbf{E}}^\top$, computed in the low-rank order so it never forms the $N \times N$ matrix and costs $\mathcal{O}(Nr)$; the trust weight λ is learned. Because $\mathbf{E}$ receives gradient from the task loss ($\partial\mathcal{L}/\partial\mathbf{E} \neq 0$), the graph is shaped to be discriminative rather than merely variance- or centrality-maximal, and its r latent factors act as learned gene programs that denoise each gene toward its program’s consensus. This graph can be initialised randomly or *warm-started* from the biological graph (top- r eigenmodes of the co-expression / Reactome operator); we compare both. As Sec. 4 shows, this learned graph is the one routing variant that decisively helps.

Training Objective

We minimise a composite loss $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{marker}} + \gamma\mathcal{L}_{\text{div}} + \zeta\mathcal{L}_z + \eta\mathcal{L}_{\text{bal}}$, where $\mathcal{L}_{\text{task}}$ is class-weighted cross-entropy (inverse-frequency weights on the train split); $\mathcal{L}_{\text{marker}}$ is the cross-entropy of an auxiliary linear classifier fed only the pre-recursion cluster tokens, forcing the marker head to select task-sufficient genes; $\mathcal{L}_{\text{div}}$ is the off-

diagonal energy of the normalised marker Gram matrix (prevents marker collapse); $\mathcal{L}_z$ is the router logit z -loss; and $\mathcal{L}_{\text{bal}}$ is the Switch load-balancing term (Shazeer et al. 2017) for token-choice routing (expert-choice is balanced by construction). Unless noted $\lambda=0.1, \gamma=0.05, \zeta=10^{-3}, \eta=0.1$; τ anneals geometrically from 1 to 0.1 and each router query is peak-initialised.

4 Experiments

Setup

We evaluate on all 11 datasets. The 8 **genomap single-cell** suites (Islam and Xing 2023) – Baron, Lung, Muraro, Oesophagus, Segerstolpe, Spleen, T-cell and Xin – span an easy-to-hard range of scRNA-seq cell-recognition tasks. The 3 **Reactome/P-NET multi-omics cohorts** – prostate, BLCA and STAD – combine mutation, copy-number and expression channels through fixed Reactome pathway tokens, testing whether the same design transfers beyond single cell; there is no TCGA bulk data. We follow the genomap-paper protocol: each dataset’s train/test split, AdamW with learning rate 10^{-3} and weight decay 10^{-5} , batch size 128, up to 150 epochs with early stopping on a held-out validation slice, per-gene z -scoring fit on the train split. Unless noted, $d=96, M=128$ markers, recursion depth $K=4$. The biology-informed router uses $k=16$ neighbours and an annealed $\beta_0=1$; the learned routing graph is trained end-to-end. Numbers are the mean $\pm$ std over 10 seeds for the routing ablation (Table 1) and over 3 seeds for the architecture, token and calibration sweeps, all with the hard arg-max marker panel at inference.

Roadmap. Five result tables settle five questions. *Does biology help routing?* Table 1: a *learned* routing graph is the decisive positive, while a *fixed* hand-built biology prior does not beat a degree-matched random-graph control. *Does the architecture cost accuracy?* Table 2: the vanilla-to-MoR ladder preserves accuracy while cutting parameters and compute. *How small can the parameters get?* Table 3: an exact $K \times$ reduction from weight sharing. *How few tokens suffice?* Table 4: a few dozen to a few hundred markers recover most full-gene accuracy. *Do the priors improve calibration?* Table 5: no. Every table is multi-seed over the full 11-dataset suite.

Main Results: Learned Routing Is the Decisive Positive

Our central routing experiment holds the architecture fixed and varies only how the depth router’s gene-gene graph is formed, under otherwise identical training. We compare five settings: *none* (a data-only router, no graph); a degree-matched *random-graph* control; a *fixed biology* prior (a hand-built graph: genomap co-expression centrality on single cell, curated Reactome centrality on P-NET); a *learned* graph trained end-to-end from the expression data and labels (randomly initialised, no explicit biological prior); and *learned_{bio}*, the same learned graph but *warm-started* from the biological graph (top- r eigenvectors of the co-expression / Reactome operator) and then refined end-to-end. Note the learned graph is not “biology-free”: it discovers gene-gene structure from the biological data itself; what it lacks is an *externally supplied* biological prior. Table 1 reports test macro-F1 for all five settings on every dataset.

The learned graph is the clear winner: it lifts mean single-cell macro-F1 to 70.8% (85.4% accuracy), +11.7 points over the no-prior router (59.0%) and well above the random-graph control (57.5%), and it is best on nearly every suite. The *fixed* biology prior, by contrast, is the honest negative: at 42.4% single-cell macro-F1 it sits *below* both the no-prior and random-graph baselines and collapses outright on several suites (Muraro, Segerstolpe, Xin), because a rigid, annealed hand-built graph pins the router to a bad operating point it cannot leave. The decisive comparison for a reviewer is *learned_{bio}* versus the general learned graph, which isolates whether an *explicit* biological graph adds anything on top of end-to-end learning: the bio-initialised graph matches the general (randomly-initialised) learned graph within noise (71.6% vs. 70.8%), so the model recovers the useful biological structure from the data on its own and an explicit biological warm-start is neither necessary nor harmful. The lesson is that biology helps routing when the graph is *learned from data* rather than imposed rigidly a priori; supplying that graph as a warm-start, rather than a frozen prior, is what removes the collapse of the fixed-biology setting.

The Vanilla-to-MoR Ladder Preserves Accuracy

Table 2 reads the architecture as a ladder from a vanilla transformer (K independent layers) to our expert-choice MoR, pooling accuracy and macro-F1 over all 11 datasets. Tying the K independent layers into one weight-shared block makes

Dataset	None	Random	Biology	Learned	Learned _{bio}
<i>Single-cell (genomap)</i>					
Baron	59.7 ± 4.7	60.6 ± 4.0	69.7 ± 2.5	79.1 ± 2.8	82.7 ± 3.5
Lung	71.6 ± 1.3	69.6 ± 2.5	72.3 ± 2.2	79.1 ± 1.2	79.3 ± 0.8
Muraro	69.7 ± 4.9	67.4 ± 6.4	2.1 ± 0.0	75.3 ± 6.4	76.3 ± 5.7
Oeso.	56.7 ± 3.4	55.4 ± 3.2	59.0 ± 3.4	69.6 ± 1.4	70.0 ± 1.1
Seger.	56.5 ± 7.6	49.5 ± 9.9	0.1 ± 0.0	71.4 ± 5.3	69.7 ± 5.5
Spleen	48.9 ± 3.0	48.1 ± 2.6	56.7 ± 2.2	58.2 ± 0.9	58.7 ± 1.1
T-cell	49.9 ± 4.0	49.6 ± 3.0	61.0 ± 3.6	68.9 ± 1.0	68.8 ± 1.0
Xin	59.3 ± 5.8	59.6 ± 9.4	18.6 ± 0.0	64.7 ± 9.4	67.4 ± 6.7
<i>Multi-omics (Reactome/P-NET)</i>					
Prostate	60.3 ± 2.9	64.1 ± 4.4	65.2 ± 3.3	66.0 ± 3.7	62.6 ± 5.1
BLCA	44.4 ± 5.0	44.0 ± 6.0	43.7 ± 5.9	49.2 ± 4.3	49.5 ± 5.8
STAD	51.6 ± 5.9	49.1 ± 5.1	49.2 ± 5.3	52.0 ± 5.8	50.2 ± 4.3
Mean macro-F1	57.1	56.1	45.2	66.7	66.8
Mean accuracy	71.1	69.0	58.1	77.9	77.9

Table 1: **Biology-informed routing: fixed vs. learned vs. bio-initialised.** Test macro-F1 (mean±std over 10 seeds) for five depth-router graph sources, holding the architecture fixed. *Biology* is a fixed hand-built graph; *Learned* is trained end-to-end from data (random init, no explicit prior); *Learned_{bio}* is that learned graph warm-started from the biological graph. The learned graphs win; the fixed prior sits at or below its random-graph control. Bottom rows: mean over all 11 datasets.

Configuration	Cost (design)		Suite mean (11)	
	Params	FLOPs	Accuracy	Macro-F1
Vanilla transformer (K independent layers)	4×	1.00×	72.6 ± 12.0	56.8 ± 10.2
Recursive, fixed depth	1×	1.00×	73.3 ± 13.7	58.2 ± 10.0
Adaptive MoR, token-choice	1×	0.56×	73.8 ± 14.2	59.7 ± 11.9
Adaptive MoR, expert-choice (ours)	1×	0.62×	72.2 ± 13.7	56.1 ± 10.9

Table 2: **Efficiency ladder.** From a vanilla transformer to expert-choice MoR: design-time parameter and FLOP cost, and accuracy / macro-F1 pooled over all 11 datasets. Weight sharing removes the 4×

parameter count independent of depth (the 4×

Parameter Efficiency Is Architectural

Table 3 contrasts the transformer-stack parameters of our shared recursion against an equivalent stack of K independent layers at matched width. The saving is present *before any training*: one shared block uses $1/K$ of the parameters of K independent layers, so at $K=4$ the shared model uses 4×

Depth K	Shared (ours)	Independent	Reduction
1	74,880	74,880	$1\times$
2	74,880	149,760	$2\times$
3	74,880	224,640	$3\times$
4	74,880	299,520	$4\times$
6	74,880	449,280	$6\times$
8	74,880	599,040	$8\times$

Table 3: **Parameter reduction.** One shared block versus K independent layers at matched width: an exact $K\times$ reduction, present before any training.

Dataset	$M=16$	$M=32$	$M=64$	$M=128$	$M=256$
<i>Single-cell (genomap)</i>					
Baron	33.7 ± 7.1	38.5 ± 2.1	53.9 ± 1.9	62.6 ± 5.1	66.9 ± 3.4
Lung	30.8 ± 5.9	38.1 ± 1.7	61.5 ± 1.9	71.4 ± 1.1	78.4 ± 0.8
Muraro	55.8 ± 3.6	59.4 ± 1.3	67.9 ± 5.6	68.5 ± 4.1	76.2 ± 5.2
Oeso.	23.1 ± 1.3	30.2 ± 3.9	42.0 ± 2.4	53.9 ± 4.4	63.2 ± 2.4
Seger.	41.2 ± 12.1	45.3 ± 5.6	59.3 ± 4.0	59.1 ± 1.6	55.4 ± 7.0
Spleen	18.7 ± 2.4	26.8 ± 3.0	38.1 ± 2.3	48.1 ± 5.0	59.4 ± 0.9
T-cell	25.5 ± 3.2	36.6 ± 5.4	43.1 ± 5.2	47.4 ± 2.2	61.5 ± 1.7
Xin	50.5 ± 5.9	46.2 ± 1.6	60.8 ± 5.6	64.1 ± 3.3	72.4 ± 11.7
<i>Multi-omics (Reactome/P-NET)</i>					
Prostate	74.3 ± 3.0	74.7 ± 2.7	78.0 ± 3.0	76.7 ± 2.6	77.3 ± 0.2
BLCA	49.8 ± 2.2	46.0 ± 3.1	49.3 ± 2.5	48.1 ± 6.8	55.7 ± 3.5
STAD	54.2 ± 3.1	51.0 ± 2.2	44.0 ± 5.6	51.2 ± 2.8	49.4 ± 4.1
Mean macro-F1	41.6	44.8	54.4	59.2	65.1

Table 4: **Marker-token budget.** Macro-F1 (mean $\pm$ std over 3 seeds) as the marker budget M shrinks from 256 to 16. A few dozen to a few hundred tokens recover most of the full-gene accuracy. Bottom row: mean over all 11 datasets.

A Few Marker Tokens Suffice

Does the marker interface throw away signal? Table 4 sweeps the marker-token budget M and tracks macro-F1 as it shrinks. Most of the full-gene signal survives with only a few dozen to a few hundred interpretable tokens, so the $\mathcal{O}(N^2) \rightarrow \mathcal{O}(M^2)$ compression that motivates the architecture costs little accuracy. The learned cross-attention router attends softly over every gene during training and collapses to a hard arg-max panel at evaluation, yielding the interpretable recursion-depth ranking that fixed panels cannot provide.

The Priors Do Not Improve Uncertainty Either

Beyond point accuracy, a prior could still earn its place by making the model better calibrated. Table 5 reports log-probability uncertainty – negative log-likelihood and expected calibration error (lower is better) and AUROC – for the same configurations, averaged over all 11 datasets. Crucially, this includes the *learned* routing graph, the accuracy winner of Table 1: it too is *no better calibrated* than the vanilla (K -independent) transformer (NLL 1.31 vs. 1.30; ECE 0.092 vs. 0.088, both higher/worse), as are the fixed-biology router and the adaptive-depth model. So the learned graph’s decisive accuracy gain does *not* carry over to calibration: better predictions here do not mean better-calibrated ones, and no routing prior – learned or fixed – improves uncertainty. This isolates the paper’s positive contributions to the learned routing accuracy gain and the efficiency results.

Configuration	NLL $\downarrow$		ECE $\downarrow$		AUROC $\uparrow$
	raw	+ T	raw	+ T	
Vanilla (K independent layers)	1.30 ± 1.26	0.73 ± 0.22	0.088 ± 0.071	0.073 ± 0.074	0.723
Shared recursion, no prior	1.29 ± 1.20	0.72 ± 0.22	0.083 ± 0.070	0.076 ± 0.070	0.725
Fixed-depth recursion	1.36 ± 1.56	0.67 ± 0.24	0.090 ± 0.082	0.061 ± 0.050	0.731
Single pass ($K=1$)	2.71 ± 3.31	0.76 ± 0.20	0.145 ± 0.165	0.073 ± 0.061	0.712
Adaptive MoR	1.45 ± 1.28	0.72 ± 0.22	0.093 ± 0.083	0.077 ± 0.070	0.728
Fixed biology prior	1.79 ± 1.36	0.75 ± 0.25	0.070 ± 0.091	0.041 ± 0.048	0.678
Learned routing graph	1.31 ± 1.20	0.67 ± 0.29	0.092 ± 0.049	0.106 ± 0.136	0.704

Table 5: **Calibration, raw and temperature-scaled.** NLL and ECE per configuration (mean over all 11 datasets, 3 seeds), each shown raw and after temperature scaling (+ T : one scalar fit on validation, which leaves accuracy/macro-F1 exactly unchanged). No routing prior improves raw calibration – including the accuracy-winning learned graph – but temperature scaling sharply reduces NLL for every configuration (ECE improves more modestly, limited by the small per-cohort validation splits), at no accuracy cost.

Recovering calibration without touching accuracy. The miscalibration is a property of the raw softmax, not of the routing, and it is largely removed by a standard post-hoc fix. The “+ T ” columns of Table 5 apply *temperature scaling* (Guo et al. 2017): a single scalar T is fit on the validation set and divides the logits before the softmax. Because dividing all logits by one positive scalar is monotonic, the arg-max – and hence every accuracy and macro-F1 number in this paper – is *exactly unchanged*; only confidence is recalibrated. It sharply reduces the negative log-likelihood of every configuration, including the accuracy-winning learned graph (NLL $1.31 \rightarrow 0.67$, a $\sim 46\%$ reduction, comparable to the vanilla transformer’s $1.30 \rightarrow 0.73$). ECE improves more modestly and less uniformly, because the per-cohort T is fit on small validation splits. Temperature scaling helps all configurations alike, so it does not change the *comparison* – the priors still do not *differentially* improve calibration – but it makes the deployed model both accurate (via the learned graph) and far better scored.

5 Discussion and Limitations

SMART shows that biological inductive bias and parameter-efficient recursion can be co-designed: the same mechanism that makes the model small (weight sharing, compression) also makes it interpretable (markers, recursion depth), and a label-free gene-gene interaction prior can be folded directly into the routing decision without leaking labels or adding parameters. We scope the claims to what the evidence supports. (i) *The routing prior is evaluated honestly.* A benefit is credited only where a graph separates from a degree-matched random-graph control (Sec. 4); the *learned* graph does so, the *fixed* biology prior does not, and we report that comparison as the runs deliver it. (ii) *Learned beats hand-built.* A fixed centrality prior injects no usable structure once the router can learn its own graph, and even hurts on some suites; biology helps routing only when the graph is learned from data. (iii) *Adaptive routing buys compute, not accuracy.* Across the suite the routing and sharing variants cluster within run-to-run noise, so the architecture’s benefit is efficiency and the interpretable depth signal rather than an accuracy gain from depth itself. (iv) *Calibration needs a separate fix.* No routing variant improves probability calibration, and the accuracy-

winning learned graph is no better calibrated than a vanilla transformer; calibration is an orthogonal axis, and a standard post-hoc temperature scaling restores it at no accuracy cost (Table 5). (v) *Richer priors and broader data*. Pathway-membership or regulatory-network priors, the optional logit-Laplacian smoothing of Appendix C, and larger single-cell atlases are the natural next steps to test where a learned graph bites hardest.

6 Conclusion

We presented SMART, a recursive marker-guided transformer whose central novelty is a *biology-informed router* that folds a gene-gene network-centrality prior into a Mixture-of-Recursions depth decision, so biology shapes where the model spends computation rather than only how its results are read. Across 11 single-cell and multi-omics datasets we find that a *learned* routing graph is the decisive positive (single-cell macro-F1 70.8%, +11.7 over a no-prior router), while a *fixed* hand-built biology prior does not separate from a random-graph control. By learning marker genes, compressing around them, and sharing one block across recursive refinement, SMART classifies with several times fewer transformer parameters than independent layers, a $\sim 38\%$ compute saving, and an interpretable compute-allocated recursion-depth signal. The complete pipeline, including all experiments and this paper, regenerates from a single command.

7 Broader Impact and Ethics Statement

Positive applications. SMART targets cell-type annotation and bulk-omics subtyping with an order-of-magnitude fewer transformer parameters and an explicit, auditable marker-gene and recursion-depth signal; cheaper, more interpretable models lower the barrier for biological discovery and make automated annotation easier to scrutinise before it informs downstream science. *Risks and mitigations.* Cell-type and subtype predictions are research tools, not clinical decisions: deployed naively on a population or tissue absent from training they can be confidently wrong, as our hard, low-accuracy suites show (e.g. the Segerstolpe and STAD cases). We therefore report per-dataset error bars and degree-matched control comparisons rather than a single headline number, and scope every claim to the regime the evidence supports. The learned marker panels are interpretable and should be inspected for confounds (batch, donor, ambient RNA) before any biological conclusion is drawn. *Data and privacy.* All datasets are public, de-identified single-cell and bulk-omics resources used under their original licenses; we add no re-identifying information and release no individual-level data. Genomic data is inherently sensitive, and any extension of this method to non-public cohorts should pass the corresponding IRB and data-governance review. *Compute footprint.* The efficiency that motivates the method also bounds its environmental cost: every run in this paper fits on a single GPU, and weight sharing plus marker compression cut both parameters and attention FLOPs, reducing rather than inflating training cost. *Reproducibility and tooling disclosure.* The full pipeline—data preparation, training, ablations, tables and figures—regenerates from a single command, and auto-

mated tooling was used to assist with code and manuscript preparation; all reported numbers trace to committed result files.

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A Dataset Details

We use 8 genomap single-cell datasets (Islam and Xing 2023) converted to plain CSV (expression + labels + stratified split) – Baron, Lung, Muraro, Oesophagus, Segerstolpe, Spleen, T-cell and Xin – and 3 Reactome/P-NET multi-omics cohorts (prostate, BLCA, STAD) whose fixed Reactome pathway tokens pool mutation, copy-number and expression channels. Per-dataset sample, feature and class counts are read directly from the result files and summarised in Table 6.

Dataset	Modality	Samples	Features	Classes
Baron	single-cell	8,569	2,000	13
Lung	single-cell	57,020	1,089	25
Muraro	single-cell	2,122	2,000	9
Oeso.	single-cell	87,947	1,089	18
Seger.	single-cell	2,127	2,000	11
Spleen	single-cell	94,129	1,089	28
T-cell	single-cell	24,007	1,089	7
Xin	single-cell	1,492	2,000	4
Prostate	multi-omics	1,011	1,223	2
BLCA	multi-omics	404	1,258	2
STAD	multi-omics	414	1,258	2

Table 6: The 11 datasets used throughout: 8 genomap single-cell suites and 3 Reactome/P-NET multi-omics cohorts. Counts are read directly from the result files.

B Effective-FLOPs Accounting

The compute numbers report per-sample FLOPs of the recursive transformer stack, the only component routing changes. One application of the shared block to a tokens costs $\phi(a) = 4a^2d + 4ad d_{\text{ff}}$; the nominal fixed-depth cost is $\Phi_{\text{nom}} = K\phi(M)$ and the effective cost sums one block over

the tokens active at each step, $\Phi_{\text{eff}} = \sum_{t=1}^K \phi(a_t)$, where a_t is the mean number of markers the expert-choice funnel keeps at step t . Gene embedding and marker selection are $\mathcal{O}(Nd)$, identical across routing modes, and excluded from this stack-level comparison.

C Theoretical Foundation of the Router

This appendix gives the mathematical and biological grounding for SMART’s biology-informed router.

Routing as conditional computation. The router implements *conditional computation*: a learned policy routes each token to a token-specific amount of compute, the gating principle of sparsely-gated mixtures of experts (Shazeer et al. 2017), reused by Mixture-of-Depths (Raposo et al. 2024) and Mixture-of-Recursions (Bae et al. 2025); the “experts” here are recursion *depths* of one shared block, which couples adaptive computation (Graves 2016) to weight sharing.

The differentiable handle. The discrete top- k has zero gradient almost everywhere, so SMART keeps the soft probability of the chosen route as a multiplicative *gate* on the block output, $\mathbf{o}_m = g_m f_\theta(\mathbf{h}_m)$ with $g_m = \sigma(\tilde{r}_m)$. Because g_m is smooth in $\mathbf{w}_r$, the chain $\mathcal{L} \leftarrow \mathbf{o}_m \leftarrow g_m \leftarrow \mathbf{w}_r$ is unbroken: the hard choice routes, the soft gate carries the gradient. The biological prior of Eq. (1) is an additive constant in this logit, so it shifts the decision without breaking this path.

Biological foundation of the prior. Two facts from single-cell biology motivate the prior. A small set of *marker* genes carries most discriminative signal (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023), so compute should be allocated unevenly. And gene co-expression networks are approximately scale-free: a few high-degree *hub* genes (master regulators) exert outsized influence. We operationalise “hub” as eigenvector centrality on the genomap co-expression graph $\mathbf{W}$: the leading eigenvector $\mathbf{v}$ of $\mathbf{W}\mathbf{v} = \lambda\mathbf{v}$ scores each gene by the centrality of its neighbours, recursively, and we z -score it to form π .

Empirical-Bayes reading. Equation (1) is a log-linear prior on the routing decision, with $\beta_t \pi_m$ a Gaussian-like prior mean and $\beta_t = \beta_0(1 - \text{progress})$ a shrinkage strength that decays as data evidence accumulates: prior-dominated when the likelihood is uninformative (early training), data-dominated later.

Leakage safety. $\mathbf{W}$ is computed from training-split *expression only*; no cell-type label enters it. The prior therefore injects network topology, not the answer, which is why a gene-discovery claim under this prior is not circular, unlike a prior built from curated cell-type marker lists.

Optional pathway-graph smoothing. The additive bias treats genes independently. Co-pathway genes should route coherently, which one obtains by smoothing the logits over $\mathbf{W}$ with the normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, $\hat{\mathbf{r}}^{(t)} = \tilde{\mathbf{r}}^{(t)} - \gamma \mathbf{L}\tilde{\mathbf{r}}^{(t)}$ (graph-Laplacian regularisation): a gene borrows routing strength from its network neighbours. This is one sparse matrix-vector product per step and adds no transformer parameters; we expose it as an option and leave its evaluation to future work.